Time Series Econometrics for Finance

Assignment 2 — Solutions

A. Empirical Exercise: Unit-Root Test on Log Real GNP

1) Series Selection and Plot

The series chosen is **Real GNP (RGNP)** from the Nelson-Plosser dataset. After removing the leading zero observations (1860–1908), the usable annual sample runs from **1909 to 1970** (T = 62 observations). The log transformation $y_t = \ln(RGNP_t)$ is applied as instructed.

The upper panel of Figure 1 shows a clear and persistent *upward trend* with no obvious tendency to revert to a fixed mean — a first visual indication that the series may be non-stationary. The lower panel shows the first difference (approximately annual growth rates), which fluctuates around zero with no apparent trend, consistent with a stationary process after differencing.

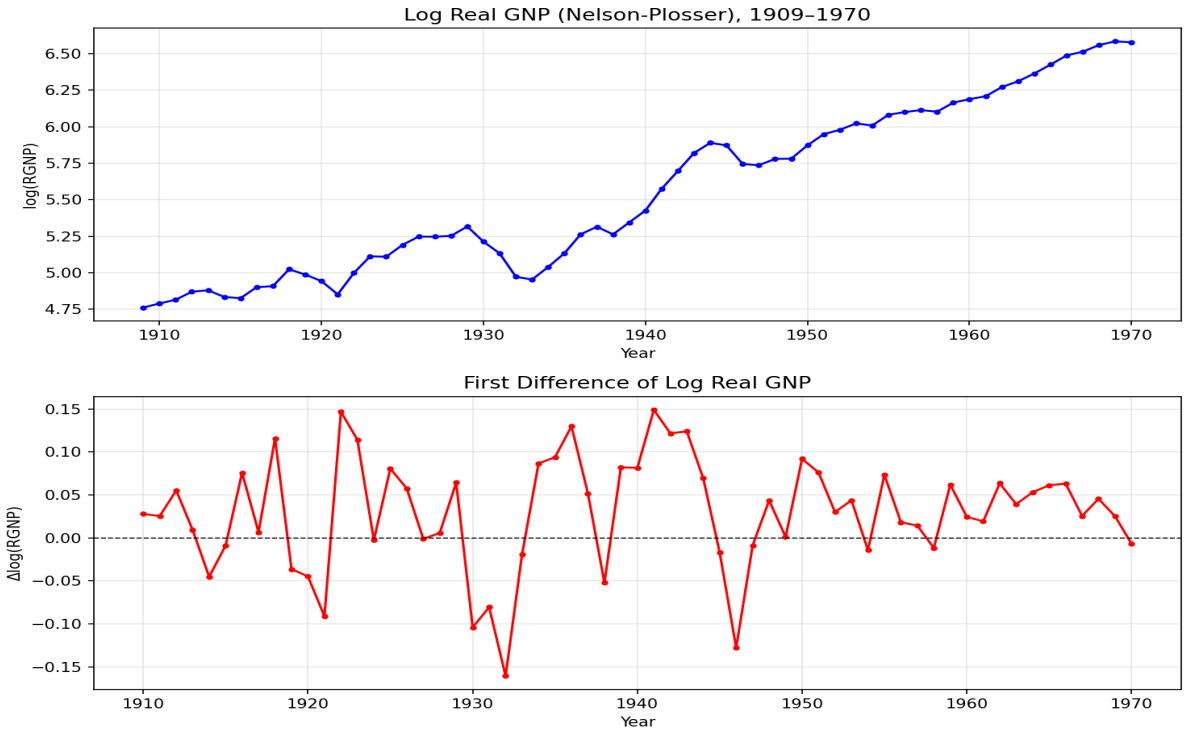

Figure 1. Log Real GNP (top) and its first difference (bottom), 1909–1970.

2) Data Preparation

Observations with zero values (years 1860–1908) are dropped, leaving a clean balanced sample. The series is then log-transformed: $y_t = \ln(RGNP_t)$. RGNP is not the Bond Yield series, so the log transformation applies.

3) Unit-Root Testing Format

We follow the **top-down approach** described in the Lecture Notes (pp. 138–141), incorporating prior information from the visual inspection of the series:

Step 1. Since the log level clearly trends upward, we begin with the most general ADF specification — **Model 3** with a constant and a deterministic time trend:

$$\Delta y_t = a_0 + a_2 t + \gamma y_{t-1} + b_1 \Delta y_{t-1} + \varepsilon_t$$

Step 2. The lag order is selected by BIC over lags 0–4 applied to the unrestricted model. BIC selects **p = 1** lagged difference.

Step 3. The key test is the **joint F-test (φ_3)**, which tests H_0: $\gamma = 0$ *and* $a_2 = 0$ simultaneously (unit root with no deterministic trend). The restricted model under H_0 is:

$$\Delta y_t = a_0 + b_1 \Delta y_{t-1} + \varepsilon_t \quad \text{(random walk with drift)}$$

The F-statistic is compared to the Dickey-Fuller φ_3 critical values (5%: 6.49, 10%: 5.47) from Fuller (1976). Standard t-table critical values are *not* applicable — the null distribution is non-standard under a unit root.

The individual ADF t-statistic on γ is also reported for completeness and compared to the DF critical values for the model with constant and trend (5%: –3.49, 1%: –4.12).

4) Results and Interpretation

Test	Statistic	CV 1%	CV 5%	CV 10%	Decision
ADF t-stat (ct, p=1)	-2.9939	-4.118	-3.486	-3.171	Fail to reject H■
φ3 F-stat (ct, p=1)	4.619	—	6.49	5.47	Fail to reject H■
ADF t-stat (Δy, n, p=0)	-4.692	-2.604	-1.946	-1.613	Reject H■

Table 1. ADF test results for log(RGNP), 1909–1970.

Interpretation:

Both the ADF t-statistic (–2.994) and the φ_3 F-statistic (4.619) fall well above their respective 5% critical values of –3.49 and 6.49. We therefore **fail to reject the null hypothesis of a unit root** at any conventional significance level. Moreover, the φ_3 test jointly rejects neither the unit root nor the deterministic trend, confirming that no significant trend is present beyond the stochastic one.

In contrast, the ADF test applied to the **first difference** Δy_t yields a statistic of –4.692, which comfortably exceeds the 1% critical value of –2.604. The first difference is strongly stationary.

Conclusion: log(RGNP) is an I(1) process — a difference-stationary series driven by a stochastic trend (random walk with drift). This replicates the original finding of Nelson and Plosser (1982), who showed that most US macro series cannot be distinguished from unit-root processes, implying that permanent shocks have lasting effects on the level of real output.

B.1 Computer Exercise: Random Walk Monte Carlo Simulation

Setup. We simulate N = 100,000 independent sequences of the pure random walk

$$x_t = x_{t-1} + u_t, \quad u_t \sim N(0,1), \quad x_0 = 0, \quad T = 81$$

For each simulated series, three OLS models are estimated:

Model	Specification	Deterministics
1	x■ = ρ x■■■ + u■	None
2	x■ = c + ρ x■■■ + u■	Constant
3	x■ = c + δt + ρ x■■■ + u■	Constant + Trend

(a) Percentiles of ρ■ and t-statistics

	Mean ρ■	Std ρ■	14th pct	86th pct
Model 1 (no constant)	0.9787	0.0380	0.9442	1.0091
Model 2 (constant)	0.9359	0.0525	0.8831	0.9847
Model 3 (const+trend)	0.8794	0.0681	0.8076	0.9465

Table 2. Distribution of OLS estimates of ρ under the unit-root DGP.

	Mean t-stat	Std t-stat	14th pct	86th pct
Model 1 (no constant)	-0.421	0.990	-1.442	+0.672
Model 2 (constant)	-1.550	0.867	-2.448	-0.625
Model 3 (const+trend)	-2.219	0.792	-3.046	-1.403

Table 3. Distribution of t-statistics (ρ■–1)/se(ρ■) under H■: ρ=1.

(b) Distributions of ρ■ and t-statistics

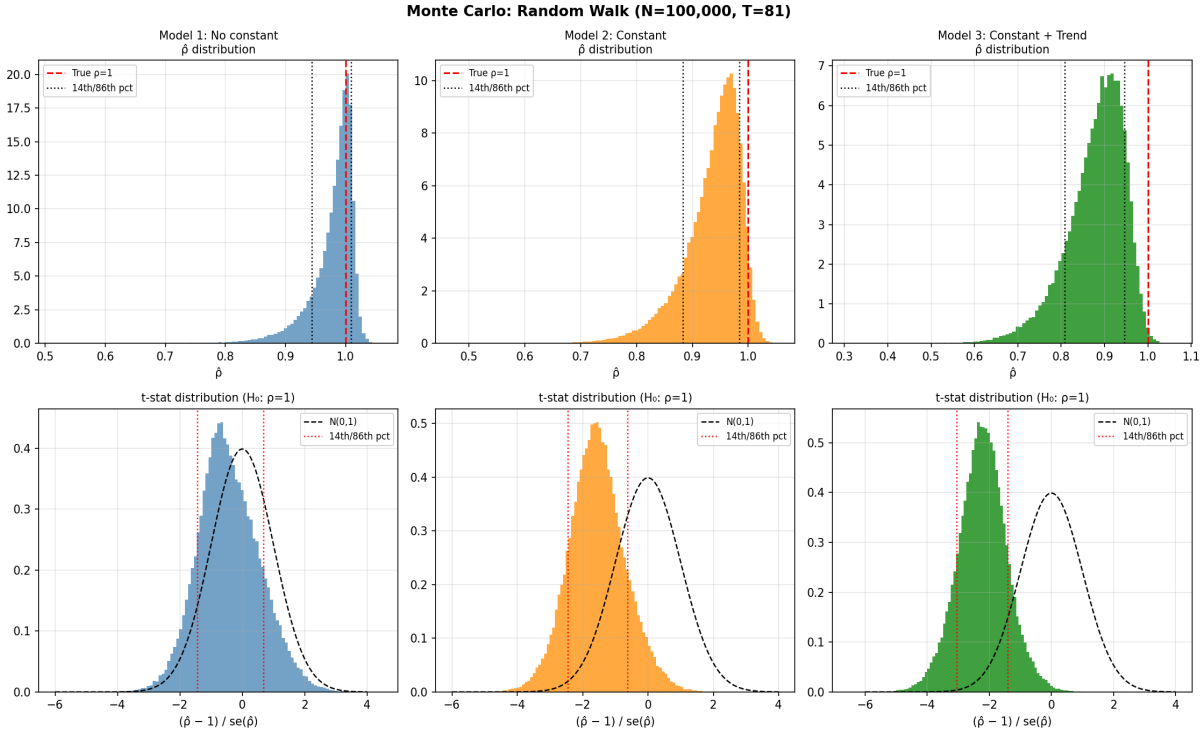

Figure 2. Top row: distribution of ρ■ for each model (red dashed = true ρ=1, dotted = 14th/86th percentiles). Bottom row: t-statistic distribution vs N(0,1) reference.

(c) Comments

Downward bias of ρ■. Even though the true value is ρ = 1, the OLS estimator is systematically biased downward. The bias worsens as more deterministic components are included: Model 1 has a mean ρ■ ≈ 0.979, while Model 3 has mean ρ■ ≈ 0.879. This reflects the fact that including a constant or trend "absorbs" variation from x_{t-1}, pulling the estimated coefficient toward zero.

Non-standard t-distribution. The t-statistics are *not* normally distributed under the unit-root null. They are left-skewed and shifted well below zero — especially when deterministic terms are present. Model 3 has a mean t-stat of –2.22, far from the N(0,1) mean of zero. This is the Dickey-Fuller distribution.

Critical implication for inference. If one incorrectly used standard t-table critical values (–1.645 at 5%) to test H_0: ρ = 1, one would massively over-reject the unit-root null. The simulated 14th and 86th percentiles give the empirical DF critical values, confirming that the correct 5% left-tail cutoff is around –1.44 (Model 1), –2.45 (Model 2), and –3.05 (Model 3) — all more negative than the standard t critical values.

Role of deterministics. Adding a constant or trend under the unit-root null shifts the entire distribution of both ρ■ and the t-statistic further left. This is why Dickey-Fuller tables have three separate sets of critical values depending on the model specification chosen.

B.2 Computer Exercise: BN Decomposition and HP Filter (US Real GDP)

Data. US Real GDP (GDPC1), quarterly, 1947:Q1–2019:Q4 (T = 292 observations, truncating at 2019:Q4 to exclude COVID). The log series is $y_t = \ln(GDP_t)$.

(a) ARMA(1,0) Fit and Beveridge-Nelson Decomposition

An ARMA(1,0) model with intercept is fitted to the first differences $\Delta y_t = y_t - y_{t-1}$ by maximum likelihood. Note that BIC also selects ARMA(1,0) from models up to ARMA(2,1), confirming the specification.

$$\text{Estimated model:} \quad \Delta y_t = 0.00775 + 0.3585\Delta y_{t-1} + \varepsilon_t, \quad \sigma_{\blacksquare}^2 = 7.47 \times 10^{-5}$$

Parameter	Estimate	Std. Error	z-stat	p-value	95% CI
c (intercept)	0.00775	0.00080	9.67	0.000	[0.006, 0.009]
φ (AR coef)	0.3585	0.047	7.66	0.000	[0.267, 0.450]
σ^2	7.47×10■■	4.4×10■■	16.97	0.000	[6.6, 8.3]×10■■

Table 4. ARMA(1,0) estimation results for Δy■, 1947:Q1–2019:Q4.

The long-run mean of Δy_t is $\mu_{\blacksquare} = c_{\blacksquare}/(1-\varphi_{\blacksquare}) = 0.00775/(1-0.3585) =$ **0.01207** per quarter (~4.8% annualised), representing the steady-state quarterly growth rate of the US economy.

Beveridge-Nelson decomposition. For an ARMA(1,0) model on Δy_t, the BN transitory (cyclical) component has a closed-form expression. The BN permanent component is defined as the expected long-run level, discounting predictable future deviations from the long-run mean. Formally:

$$C^{BN}_t = -\varphi_{\blacksquare}/(1-\varphi_{\blacksquare}) \times (\Delta y_t - \mu_{\blacksquare})$$

$$\text{With } \varphi_{\blacksquare} = 0.3585: \quad C^{BN}_t = -0.558 \times (\Delta y_t - 0.01207)$$

The BN permanent component is then $\tau^{BN}_t = y_t - C^{BN}_t$. Intuitively, the cycle captures the degree to which current growth exceeds or falls short of the long-run average, scaled by the persistence of those deviations.

(b) Hodrick-Prescott Filter

The HP filter decomposes y_t into trend τ^{HP}_t and cycle $C^{HP}_t = y_t - \tau^{HP}_t$ by minimising:

$$\Sigma(y_t - \tau_t)^2 + \lambda\, \Sigma(\Delta^2\tau_t)^2$$

with smoothing parameter λ **= 1,600** (standard for quarterly data). A higher λ penalises curvature in the trend more strongly, producing a smoother permanent component.

(c) Comparison of Transitory Components

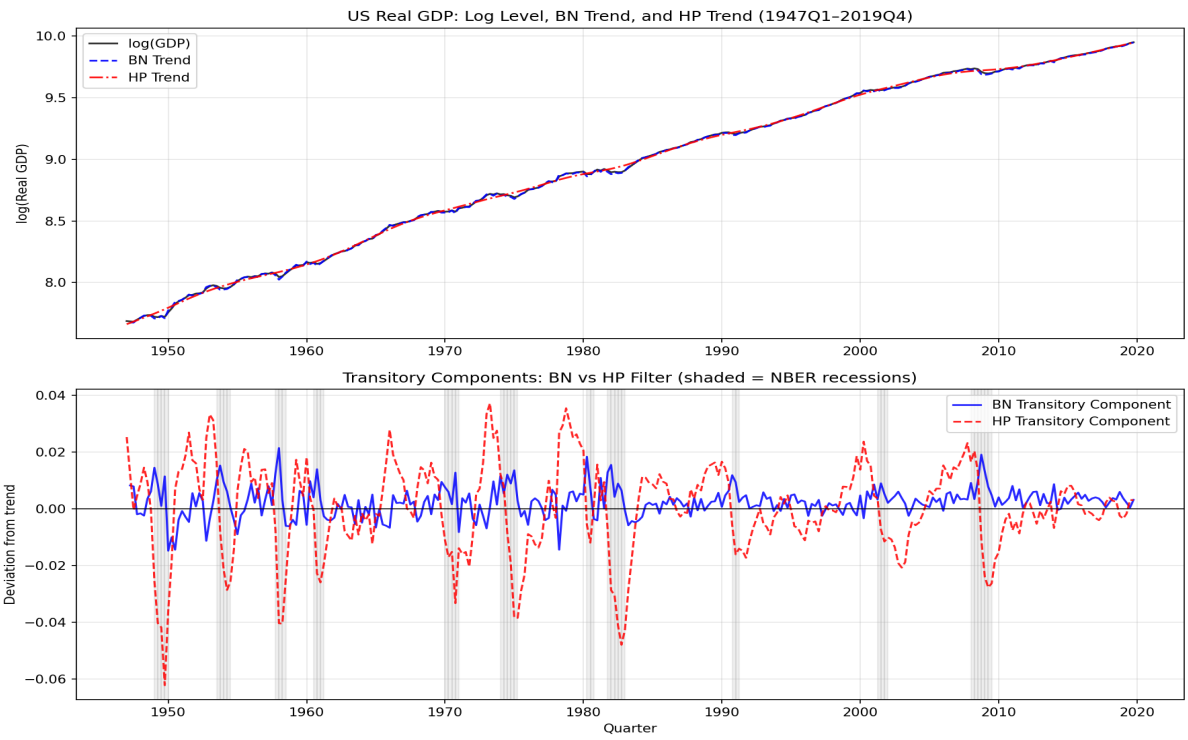

Figure 3. Top: log(GDP), BN trend and HP trend. Bottom: BN cycle (blue) vs HP cycle (red dashed), shaded areas = NBER recessions.

Statistic	BN Cycle	HP Cycle
Std. Deviation	0.0052	0.0158
Minimum	-0.0148	-0.0623
Maximum	+0.0214	+0.0372
Correlation (BN vs HP)	–0.257	—

Table 5. Summary statistics for cyclical components.

Amplitude. The HP cycle has a standard deviation of 1.58% — about three times larger than the BN cycle (0.52%). The HP filter attributes large, persistent swings in GDP (e.g., the 2008–09 Great Recession, reaching –6.2%) to the cyclical component. The BN cycle is far smaller because it incorporates predictability: since Δy_t has positive autocorrelation ($\varphi = 0.36$), above-average growth today implies above-average growth next quarter, so the permanent component rises by more than just the current increment, absorbing much of the swing.

Negative correlation. The two cycles are negatively correlated (–0.26), meaning they frequently move in opposite directions. This is a well-documented phenomenon: the BN and HP decompositions embed fundamentally different views of what constitutes the "trend." The HP trend is a smooth curve penalising curvature, while the BN trend is a martingale that jumps every period.

Recession behaviour. Both cycles turn negative during all NBER recessions (shaded grey), consistent with economic downturns reducing output below trend. The HP cycle more clearly identifies business cycle turning points and produces smoother cycles that align closely with the conventional business cycle narrative. The BN cycle is noisier and harder to interpret economically at quarterly frequency with only an AR(1), but it is theoretically grounded in the rational expectations permanent income framework.

Conclusion: The two filters agree that the US economy experienced pronounced cyclical swings around recessions, but disagree strongly on their magnitude. The choice between them depends on whether one views the trend as a smooth deterministic path (HP) or as a stochastic process that incorporates all predictable future growth (BN). Neither decomposition is universally "correct" — they answer different economic questions.